

## Problem Sets Before Test 1

*Only turn in problems that are **not** bracketed.* Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

| notation    | meaning                                                                       |
|-------------|-------------------------------------------------------------------------------|
| unbracketed | assigned problem – turn these in for grading                                  |
| ()          | helper/warm-up problem                                                        |
| []          | additional problems (you are responsible for content, but don't turn them in) |
| { }         | covers optional material                                                      |

| Due      | Task                 | Source                                  | Problems                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon 2/4  | PS 1                 | Rosen 10.1                              | (1) (3,5,7,9) types of graph <b>4</b> types of graph <b>8</b> types of graph <b>13</b> intersection graph<br><b>18</b> Fred <b>33</b> precedence graph                                                                                                                                                                                                                                                |
| Wed 2/6  | <a href="#">WW01</a> | Rosen 10.2                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| Fri 2/8  | <a href="#">WW02</a> | Rosen 10.3                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mon 2/11 | PS 2                 | Rosen 10.4                              | <b>2</b> paths <b>6</b> components <b>12</b> strong/weak connectivity <b>14</b> components<br><b>16</b> along the way <b>18</b> isomorphic? <b>20</b> isomorphic? <b>64</b> farmer puzzle <b>66</b> water puzzle<br>[1] paths [3–5] [11] strong/weak connectivity [21, 23] isomorphic?                                                                                                                |
| Fri 2/15 | PS 3                 | Rosen 10.5                              | <b>2–8 even</b> Euler paths/circuits <b>10</b> bridges<br><b>18</b> directed Euler <b>20</b> directed Euler <b>26</b> which have Euler circuits?<br><b>13–15</b> trace it<br>[1–7 odd] Euler paths/circuits [19, 21] directed Euler<br><br>Note: Your answers to homework in this class should typically include some sort of explanation. Answers like “Yes”, “No”, and “17”, are rarely sufficient. |
| Wed 2/20 | PS 4                 | Rosen 10.2<br>Rosen 10.7<br>Rosen 10.RQ | <b>22–25</b> bipartite? <b>26</b> bipartite?<br><b>2–4</b> planar <b>12</b> regions [13] regions <b>14</b> vertices<br><b>3</b> degree                                                                                                                                                                                                                                                                |
| Fri 2/22 | PS 5                 | Rosen 10.7<br>Rosen 10.SE               | <b>6–8</b> planar <b>23–25</b> Kuratowski<br><b>3–5</b> isomorphic?                                                                                                                                                                                                                                                                                                                                   |
| Tue 2/26 | <a href="#">WW03</a> |                                         | Note Tuesday due date (at noon)                                                                                                                                                                                                                                                                                                                                                                       |
| Thu 2/28 | <a href="#">WW04</a> |                                         | Note Thursday due date (at noon)                                                                                                                                                                                                                                                                                                                                                                      |

| Due     | Task                 | Source    | Problems                                                                                                                                                      |
|---------|----------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fri 3/1 | PS 6                 | Rosen 6.1 | <b>28</b> license plates <b>48</b> bit strings <b>42</b> DNA <b>50</b> consecutive 0's<br>[29–30] license plates [32–33] strings [41] photos [56] C variables |
|         |                      | Rosen 6.2 | <b>4</b> balls <b>40</b> party                                                                                                                                |
|         |                      | Rosen 6.3 | <b>22</b> strings <b>26</b> softball                                                                                                                          |
| Tue 3/5 | <a href="#">WW05</a> |           |                                                                                                                                                               |